MODULE 32 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 1

Twenty questions, weighted to the real domain mix. Nine seconds to answer each.

TYPE Practice exam QUESTIONS 20 PAUSE 9 seconds

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How to use this

- Each question gives you **nine seconds** before the answer. Pause if you need longer.
- **Commit to an answer out loud** before the walkthrough. A guess you did not commit to teaches nothing.
- Apply module 30: read the ask, find the deciding clause, check the qualifier.
- Track your score. Under 14 out of 20 means revisit before booking.
- The real exam gives you 2 minutes per question. Nine seconds is deliberately tight.

> ON THE EXAM
>
> The nine-second pause is for recognition, not deliberation. If a question needs longer, pause the video - that is a signal about where to revise.

A Lambda function backing a checkout API is invoked in bursts after marketing emails. p50 latency is 80 ms but p99 is 2.4 seconds. CloudWatch shows Init Duration present on the slow invocations only.

Traffic is predictable: the emails go out at 09:00 each weekday. Cost matters.

What should the team do?

A Configure provisioned concurrency with scheduled scaling for the 09:00 window

B Increase the function's reserved concurrency

C Increase the function timeout so slow invocations complete

D Enable provisioned concurrency permanently at the peak level

◆ PAUSE AND DECIDE

What should the team do?

A Scheduled provisioned concurrency

Init Duration on slow invocations identifies cold starts. Predictable timing means scheduled scaling pays only for the window that needs it.

B Reserved concurrency

Reserved concurrency is a cap, not an allocation of warm environments. It does nothing about cold starts.

C Longer timeout

The invocations are completing, just slowly. A timeout is not being hit.

D Permanent provisioned concurrency

Works, and pays for warm environments 24 hours a day to serve one predictable window. The cost clause rules it out.

THE SIGNAL

Two clues: Init Duration names the cause, and 'predictable' plus 'cost matters' selects scheduled over permanent.

A DynamoDB table storing IoT readings uses deviceId as the partition key and timestamp as the sort key. One device produces 80% of all traffic. The table is in on-demand mode and reports throttling, while overall consumed capacity is low.

Reads always request the most recent hour for a single device.

What change resolves the throttling?

A Switch to provisioned capacity with auto scaling

B Add a global secondary index on timestamp

C Add a write-sharding suffix to the partition key for the high-volume device

D Enable DynamoDB Accelerator in front of the table

◆ PAUSE AND DECIDE

What change resolves the throttling?

A Provisioned with auto scaling

Changes the table's capacity model. Per-partition limits apply in both modes, so the hot partition still throttles.

B GSI on timestamp

Timestamp is a poor partition key and the reads are already efficient. A GSI does not redistribute writes to the base table.

C Write sharding on the partition key

Spreads one device's writes across several partition keys, which is the only change that addresses a per-partition limit.

D DAX

Caches reads. The throttling is on writes concentrated in one partition, which a read cache cannot help.

THE SIGNAL

'Throttling while overall capacity is low' is the hot partition signature. No capacity mode removes a per-partition limit.

A web application lets authenticated users upload files of up to 2 GB. Uploads currently pass through an EC2 fleet, which is running out of disk. The S3 bucket must remain private and users must not receive long-lived credentials.

Uploads should not traverse the application servers at all.

What is the correct approach?

A Make the bucket public for PUT and validate uploads afterwards

B Have the application return a presigned PUT URL, and upload directly using multipart

C Issue each user an IAM user with scoped credentials

D Use a CloudFront signed URL for the upload

◆ PAUSE AND DECIDE

What is the correct approach?

A Public bucket

Contradicts 'must remain private'. Any option that solves a problem by making a bucket public is essentially always wrong.

B Presigned PUT plus multipart

The bucket stays private, no long-lived credentials are issued, the bytes bypass the servers, and multipart handles 2 GB with resumable parts.

C An IAM user per user

IAM users are long-lived credentials, which the scenario forbids, and the 5,000-user quota makes it unworkable.

D CloudFront signed URL

CloudFront signed URLs are download only. They cannot sign an upload at all.

THE SIGNAL

Three requirements, one mechanism. Option D is the direction trap: the pair splits by upload versus download.

> An API Gateway REST API with a Lambda proxy integration returns 502 Bad Gateway for every request. CloudWatch shows the function completing in 30 ms with no errors logged.
>
> The handler ends by returning a plain object containing the result data.

What is the cause?

A The function has been throttled

B The integration timeout has been exceeded

C API Gateway lacks permission to invoke the function

D The response does not match the proxy integration format

◆ PAUSE AND DECIDE

What is the cause?

A Throttled

Throttling returns 429 and the function would not run. This one completes in 30 ms.

B Timeout

A timeout returns 504, not 502, and 30 ms is nowhere near 29 seconds.

C Missing permission

Produces 500 with a permission error, and the function would never be invoked.

D Wrong response shape

Proxy integration requires statusCode and a string body. A bare object leaves the gateway unable to build a response, which is exactly what 502 means.

THE SIGNAL

The function succeeded and the API still failed, which places the fault between them. That is the definition of 502.

A Lambda function consumes an SQS queue. Its timeout is 90 seconds and processing typically takes 70. The queue's visibility timeout is 30 seconds and maxReceiveCount is 4.

Downstream systems report each record created four times, and the dead-letter queue is filling steadily despite almost no Lambda errors.

What should be changed?

A Raise the visibility timeout to at least six times the function timeout

B Increase maxReceiveCount so messages are not sent to the DLQ

C Switch the queue to FIFO to prevent duplicate delivery

D Reduce the function timeout to match the visibility timeout

◆ PAUSE AND DECIDE

What should be changed?

A Raise the visibility timeout

70 seconds of work under a 30-second visibility timeout means the message reappears mid-flight and is redelivered. Four duplicates matching maxReceiveCount of 4 confirms it.

B Raise maxReceiveCount

Would produce more duplicates, not fewer. It raises the threshold without addressing why messages are redelivered.

C Switch to FIFO

FIFO suppresses duplicate sends within five minutes. It does not stop redelivery caused by a short visibility timeout.

D Reduce the function timeout

Processing takes 70 seconds, so a 30-second timeout would fail every invocation outright.

THE SIGNAL

Four duplicates and maxReceiveCount of 4 is not a coincidence. The arithmetic identifies the cause before you read the options.

A loan approval workflow must pause after automated checks until a human underwriter approves or rejects it. Underwriters typically respond within a day but occasionally take four days.

No compute should run while waiting, and the workflow must resume automatically on the decision.

How should the pause be implemented?

A A Step Functions Wait state of 96 hours followed by a check

B A Step Functions Task using .waitForTaskToken, with the token sent to the underwriter

C An Express workflow that polls an approvals table

D A Lambda function that sleeps until approval arrives

◆ PAUSE AND DECIDE

How should the pause be implemented?

A Wait 96 hours then check

Every loan waits four days even when approved in ten minutes, and it fails if an underwriter takes slightly longer.

B .waitForTaskToken

The execution pauses with no compute billed, appears as awaiting callback, and resumes the moment SendTaskSuccess arrives. Standard workflows wait up to a year.

C Express polling

Express workflows stop at five minutes, so this cannot span four days.

D A sleeping Lambda

Lambda stops at 15 minutes and bills every second of sleeping. Paying a function to wait is the anti-pattern callbacks remove.

THE SIGNAL

Three requirements: no compute while waiting, an unpredictable duration, and automatic resumption. Only the callback pattern gives all three.

> Account A must read objects from a bucket in Account B encrypted with SSE-KMS using a customer managed key. The bucket policy allows the role in Account A, and that role's identity policy permits s3:GetObject and kms:Decrypt on the key ARN.
>
> Downloads fail with AccessDenied originating from KMS.

What is missing?

A Block Public Access must be disabled on the bucket

B The key must be converted to a multi-Region key

C The role must be granted s3:ListBucket as well

D The KMS key policy in Account B must explicitly allow the role in Account A

◆ PAUSE AND DECIDE

What is missing?

A Disable Block Public Access

It blocks anonymous and public access. A named role in a named account is neither.

B Multi-Region key

Multi-Region solves cross-Region, not cross-account. Nothing here involves two Regions.

C Add ListBucket

ListBucket is needed to enumerate objects, not to read one. The error also comes from KMS, not S3.

D The key policy must allow it

KMS uniquely requires an explicit allow in the key policy. An identity policy alone is never sufficient, even in the same account.

THE SIGNAL

The error originating from KMS rather than S3 is the clue: bucket access already works, and the key is refusing separately.

A mobile app must let users sign up with email or Google, then upload images directly to Amazon S3 from the device, with each user confined to their own prefix.

The team does not want to build a user directory and does not want uploads passing through their servers.

What should the team use?

A. A Cognito user pool only, sending the ID token to S3

B. A Cognito identity pool only, federating Google

C. A user pool for sign-in, feeding an identity pool whose role uses policy variables

D. An IAM role assumed directly by the mobile app using embedded credentials

◆ PAUSE AND DECIDE

What should the team use?

A User pool only

S3 does not accept Cognito JWTs. It requires SigV4-signed requests with AWS credentials, which a user pool does not issue.

B Identity pool only

Would cover Google sign-in, and email sign-up needs a directory. Identity pools hold no users of their own.

C Both pools, with policy variables

The user pool provides the directory and Google federation; the identity pool exchanges the token for credentials, scoped per user by policy variables.

D Embedded credentials

Anything shipped to a device can be extracted. Embedding AWS credentials in a mobile app is never acceptable.

THE SIGNAL

Two requirements force both halves: email sign-up needs a directory, and direct S3 access needs AWS credentials. Neither service does both.

> Compliance requires an Amazon RDS master password to rotate every 30 days. Several applications connect to the database and none may fail during the change. The team wants to write as little code as possible.
>
> Applications read the credential at startup and cache it.

How should this be implemented?

A. A scheduled Lambda function that changes the password and updates a Parameter Store SecureString

B. Secrets Manager managed rotation on a 30-day schedule, using the alternating users strategy

C. Secrets Manager managed rotation using the single user strategy

D. A Parameter Store advanced parameter with a no-change notification after 30 days

◆ PAUSE AND DECIDE

How should this be implemented?

A. Scheduled Lambda

Rebuilds rotation by hand without staging labels or a test step, and the scenario asks for minimal code.

B. Managed rotation, alternating users

RDS is covered by managed rotation so no code is needed, and alternating users keeps the previous credential valid, so cached credentials keep working.

C. Managed rotation, single user

The strongest distractor. Single user rotation changes the password in place, leaving a window where a cached credential fails.

D. No-change notification

Notifies that the password is stale. It does not rotate anything, which is the entire requirement.

THE SIGNAL

'Applications cache the credential' plus 'none may fail' is what selects alternating users over single user. Both are managed rotation.

> A developer's role has the AdministratorAccess policy attached. They work normally in eu-west-1 but every action in us-east-1 fails with AccessDenied. The account belongs to an AWS Organization.
>
> Attaching further policies to the role changes nothing.

What is the cause?

A. A service control policy denies actions outside approved Regions

B. AdministratorAccess does not cover us-east-1 by default

C. us-east-1 must be enabled for the account

D. The role's trust policy is missing the us-east-1 endpoint

◆ PAUSE AND DECIDE

What is the cause?

A An SCP denying Regions

Administrator access failing, and more permissions not helping, is the signature of an explicit deny. An SCP is the only thing that overrides AdministratorAccess.

B Region-limited policy

AdministratorAccess is Action star on Resource star with no Region condition. It covers every Region.

C Region not enabled

us-east-1 is enabled by default in every account. Some newer Regions are opt-in; this one is not.

D Trust policy

Trust policies govern who may assume a role, not where it may act. The role is clearly already assumed.

THE SIGNAL

Adding permissions changing nothing is the defining property of an explicit deny. Inside an Organization, that means an SCP.

> An ECS task on Fargate assumes a cross-account role to read a partner's S3 bucket. The role's maximum session duration is 12 hours and the code requests 12 hours. The same code run locally with a developer's IAM user credentials succeeds.
>
> In Fargate the AssumeRole call fails with an error about the requested duration.

What explains the difference?

A Fargate tasks cannot assume cross-account roles

B The task execution role needs sts:AssumeRole rather than the task role

C The task already runs as a role, so this is role chaining, which caps sessions at one hour

D The partner account must raise its maximum session duration

◆ PAUSE AND DECIDE

What explains the difference?

A Fargate cannot assume

It can, and does routinely. Nothing about Fargate prevents cross-account assumption.

B Execution role instead

The execution role starts the container; the task role is what application code uses. This would also produce access denied, not a duration error.

C Role chaining caps at 1 hour

The task runs as its task role, so assuming another role is a second hop. Chaining caps at one hour and requesting more fails outright rather than truncating.

D Partner raises the duration

It is already 12 hours. The cap being hit is the chaining cap, which no role setting can raise.

THE SIGNAL

Two clues: the error names duration rather than permission, and it works from an IAM user but not from a role. Only chaining explains both.

> A CloudFormation template creates an IAM role and a Lambda function. The function's Role property is set with !Ref ExecutionRole. Stack creation fails with an error stating the role ARN is invalid.
>
> The role itself is created successfully.

What is wrong?

A The role needs a DependsOn attribute so it is created first

B The role's trust policy is missing the Lambda service principal

C IAM roles cannot be referenced from a Lambda resource in the same template

D Ref on an IAM role returns the role name. The property needs !GetAtt ExecutionRole.Arn

◆ PAUSE AND DECIDE

What is wrong?

A Missing DependsOn

The Ref already creates an implicit dependency, and the role was created successfully. Ordering is not the issue.

B Missing service principal

That fails at invocation with a permission error, not at creation with a message about ARN format.

C Cannot reference in one template

This is the normal pattern and works fine. Templates reference roles from functions constantly.

D Ref gives the name, use GetAtt

The Role property requires an ARN. Ref on an IAM role returns the role name, so a bare name is passed where an ARN is expected.

THE SIGNAL

The error names the problem: it says ARN. Ref returned a name, and that distinction is the single most tested CloudFormation detail.

A payments Lambda function is deployed by AWS SAM. Deployments currently replace all traffic at once, and a recent bad release caused fifteen minutes of failures before anyone noticed.

The team wants a small share of traffic on the new version first, automatic reversion if an error alarm fires, and the smallest possible template change.

What should be added?

A AutoPublishAlias and a DeploymentPreference of Canary10Percent5Minutes, with the alarm listed

B An API Gateway canary release deployment on the stage

C Provisioned concurrency so new code warms before receiving traffic

D A second function and a Route 53 weighted record between them

◆ PAUSE AND DECIDE

What should be added?

A AutoPublishAlias plus DeploymentPreference

Four lines. SAM creates a CodeDeploy deployment group, shifts 10% for five minutes, and rolls back automatically when the alarm fires.

B API Gateway canary

Shifts traffic between API deployments, not Lambda versions, and does not roll back on a CloudWatch alarm.

C Provisioned concurrency

Addresses cold starts. Warm bad code fails exactly as reliably as cold bad code.

D Two functions and weighted DNS

Builds traffic shifting by hand, with DNS TTLs making rollback slow rather than immediate.

THE SIGNAL

Option B is the layer trap: a gateway canary and Lambda alias weights are different mechanisms at different layers.

A new CodeDeploy deployment group targets EC2 instances that previously ran a different application deployed by CodeDeploy. The first deployment fails immediately, reporting that the ApplicationStop lifecycle event failed on every instance.

The new revision's scripts have been tested and work correctly.

What is the cause?

A The CodeDeploy agent is not installed

B The new appspec.yml contains invalid YAML

C The instance profile cannot read the artifact bucket

D ApplicationStop runs the previous revision's script, which is stale or broken

◆ PAUSE AND DECIDE

What is the cause?

A Agent not installed

Without the agent the deployment hangs and times out rather than reporting a specific lifecycle event failure. Naming ApplicationStop proves the agent is running.

B Invalid YAML

That fails before any lifecycle event runs, and it contradicts the statement that the new scripts work.

C Bucket permissions

That fails at DownloadBundle, which comes after ApplicationStop. The failure would name a different event.

D Previous revision's script

ApplicationStop runs from the last successfully deployed revision, because the new bundle is not on the instance yet. Old scripts are the cause.

THE SIGNAL

'Previously ran a different application deployed by CodeDeploy' is the planted clue. Those instances still hold that application's stop script.

A pipeline has a source stage, a CodeBuild stage and a CodeDeploy stage. The build reports success and all tests pass. The deploy stage then fails, reporting that the input artifact contains no deployable files.

The build writes compiled output to a build directory and the appspec file to the repository root.

What is the most likely cause?

A The CodeDeploy service role cannot read the artifact bucket

B The build and deploy stages are in different Regions

C The buildspec artifacts section does not list the build directory and the appspec file

D The build stage timed out before writing its output

◆ PAUSE AND DECIDE

What is the most likely cause?

A. Role cannot read the bucket

That produces access denied, not a report that the artifact is empty. The artifact was read successfully.

B. Different Regions

A cross-Region artifact problem fails on retrieval with a bucket error, and would have failed pipeline validation earlier.

C. Artifacts section incomplete

CodeBuild uploads only what the buildspec artifacts section lists. A green build producing an empty artifact is this, essentially every time.

D. Build timed out

A timeout fails the build. This build reported success, and CodeBuild does not upload partial output.

THE SIGNAL

'Build green, deploy finds nothing' is the single most recognisable CI/CD symptom. The build did its job and was not asked to keep anything.

> An API Gateway integration points at a Lambda function using the unqualified function ARN. A developer deploys a code fix, and production begins returning errors within seconds. Reverting requires redeploying the previous code, taking several minutes.
>
> The team wants deployments to reach production only when they choose.

What change prevents this?

A Pin the integration to a specific published version number

B Publish a version on each deployment and point the integration at an alias, updated separately

C Enable provisioned concurrency so the previous code stays warm

D Add reserved concurrency to isolate the production function

◆ PAUSE AND DECIDE

What change prevents this?

A Pin to a version

Fixes the immediacy and creates a new problem: the integration never moves, so every release means editing and redeploying the API.

B Publish a version, point at an alias

Code updates change $LATEST only, so nothing reaches production until the alias moves. Reverting is one UpdateAlias call.

C Provisioned concurrency

Addresses cold starts. It has no bearing on which code version is invoked.

D Reserved concurrency

Caps how many invocations run at once, not which version they run.

THE SIGNAL

An unqualified ARN resolves to $LATEST, so every deployment is instant and irreversible. The alias is what sits between deployment and production.

> An application writes a log line containing OrderFailed whenever an order cannot be processed. Operations want an alert when more than twenty occur in five minutes, and want the trend on a dashboard.
>
> The application code cannot be modified.

What should be configured?

A. A subscription filter to a Lambda function that counts occurrences and publishes an SNS message

B. A scheduled Logs Insights query that counts matching lines

C. Modify the application to call PutMetricData on each failure

D. A metric filter on the log group, with a CloudWatch alarm on the resulting metric

◆ PAUSE AND DECIDE

What should be configured?

A Subscription filter plus Lambda

Rebuilds a metric filter by hand, including counting and alerting, with a function to run and pay for.

B Scheduled Logs Insights

Logs Insights is interactive and billed by data scanned. Running it on a schedule is expensive and delays the alert.

C Modify the application

The scenario states the code cannot be modified.

D Metric filter plus alarm

A pattern produces a metric, the alarm watches it, and the metric charts on a dashboard with no further work.

THE SIGNAL

The question wants a number, not the events. That is the metric filter side of the pair every time.

> A support team must retrieve the X-Ray trace for a specific order when a customer reports a problem. The application attaches the order ID and the full order payload to each segment using put_metadata.
>
> Searching for the order ID in the X-Ray console returns nothing.

What change is needed?

A Record the order ID as an annotation, keeping the payload as metadata

B Increase the sampling rate to 100% so every order is traced

C Move the order ID into the segment name

D Export traces to S3 and query them with Amazon Athena

◆ PAUSE AND DECIDE

What change is needed?

A Order ID as an annotation

Annotations are indexed and filterable, so the trace becomes retrievable. The bulky payload correctly stays as metadata.

B 100% sampling

Traces every order, and the traces still cannot be searched by order ID, because metadata is not indexed.

C Order ID in the segment name

Segment names identify services, not requests. A unique name per order would create thousands of service map nodes.

D Export and query with Athena

Builds a search system to work around an indexing feature that already exists and needs one line changed.

THE SIGNAL

Only annotations are indexed. The fix is not moving everything across, but putting the searchable field in the searchable place.

> A fleet of 300 instances calls a downstream service. A brief 20-second outage became a 15-minute one. Request volume increased eightfold during the incident and stayed high after the original fault was fixed.
>
> Each instance retries failed calls five times with a fixed 500 ms delay.

What should change?

A Increase the retry count so requests eventually succeed

B Use exponential backoff with jitter, and add a retry quota or circuit breaker

C Remove retries entirely so failures surface immediately

D Add instances to spread the load

◆ PAUSE AND DECIDE

What should change?

A More retries

Directly worsens it. The eightfold volume increase is caused by retries, so adding more amplifies the storm.

B Backoff with jitter plus a quota

A fixed delay synchronises 300 clients into waves. Jitter spreads them, backoff reduces the rate, and a quota or breaker stops retrying into a sustained failure.

C No retries

Overcorrects. Transient failures that would have self-corrected now become user-visible errors.

D More instances

Increases the number of clients generating retries, making the storm larger.

THE SIGNAL

'Fixed 500 ms delay' across 300 instances is the whole diagnosis: every client retries at the same instant, five times.

> An image resizing Lambda function is configured with 128 MB of memory and takes about 6 seconds per image. The function is CPU-bound. The team wants it faster and has been told to control costs, so they have avoided raising the memory.
>
> The function runs about 50,000 times a day.

What should the team do?

A. Keep 128 MB and invoke several functions in parallel per image

B. Enable provisioned concurrency so the function is always warm

C. Run Lambda Power Tuning and expect a higher memory setting to be faster and cheaper

D. Move the workload to a continuously running EC2 instance

◆ PAUSE AND DECIDE

What should the team do?

A. Parallel functions

Each function still gets a fraction of a vCPU, so each piece is still slow. It adds orchestration to avoid a configuration change.

B. Provisioned concurrency

Removes cold start latency. The 6 seconds is execution time, so this changes nothing and adds cost.

C. Power Tuning, expect higher memory

CPU scales with memory, so a CPU-bound function at 128 MB is severely constrained. Finishing several times faster usually costs less in GB-seconds.

D. Continuous EC2

Trades per-invocation billing for an always-running instance to do roughly 83 minutes of work a day.

THE SIGNAL

The premise is the error. In serverless you pay for time, so for a CPU-bound function more memory is frequently cheaper.

Scoring this exam

- **17 or more:** comfortable. Move to practice exam 2.
- **14 to 16:** passing territory. Review the domains you dropped marks in.
- **Below 14:** revisit the relevant modules before booking.
- Which domain you missed matters more than the total. Development is 32% of the real exam.
- Re-read any answer you got right but could not justify. That is a guess, not knowledge.

> ON THE EXAM
>
> Twenty questions is a small sample, so treat the domain pattern as the signal rather than the exact number. Two more exams follow.